Railway Gauge Analysis Report

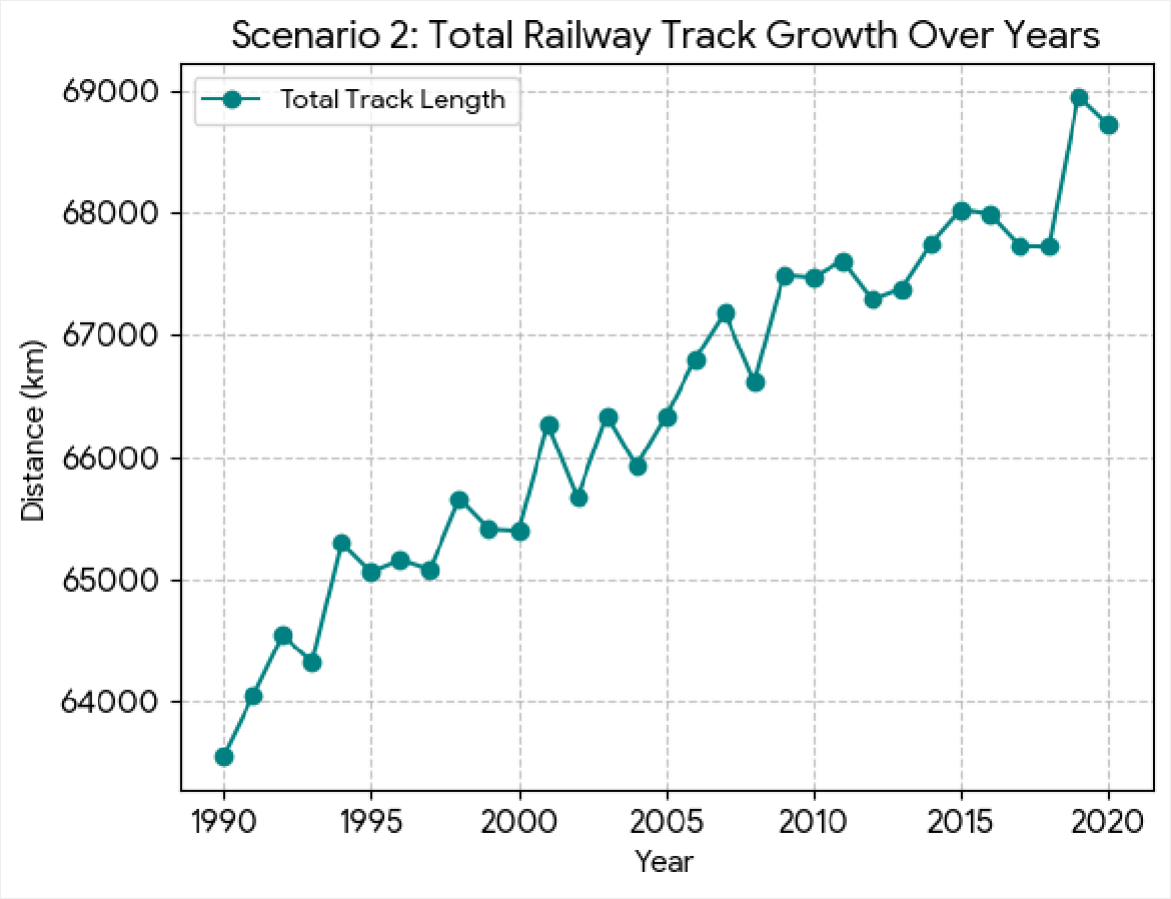
Scenario 1: Basic Data Loading & Cleaning

The dataset was loaded into a Pandas DataFrame. Missing values were replaced with 0, and all gauge-related columns were converted to numeric types for analysis.

	Year	Broad	Metre	Narrow	Total
0	1990	34681.0	24910.000000	3962.000000	63553.0
1	1991	35901.0	24131.333333	4013.666667	64046.0
2	1992	37450.0	23349.666667	3743.333333	64543.0
3	1993	37752.0	22873.000000	3698.000000	64323.0
4	1994	39493.0	22109.333333	3695.666667	65298.0

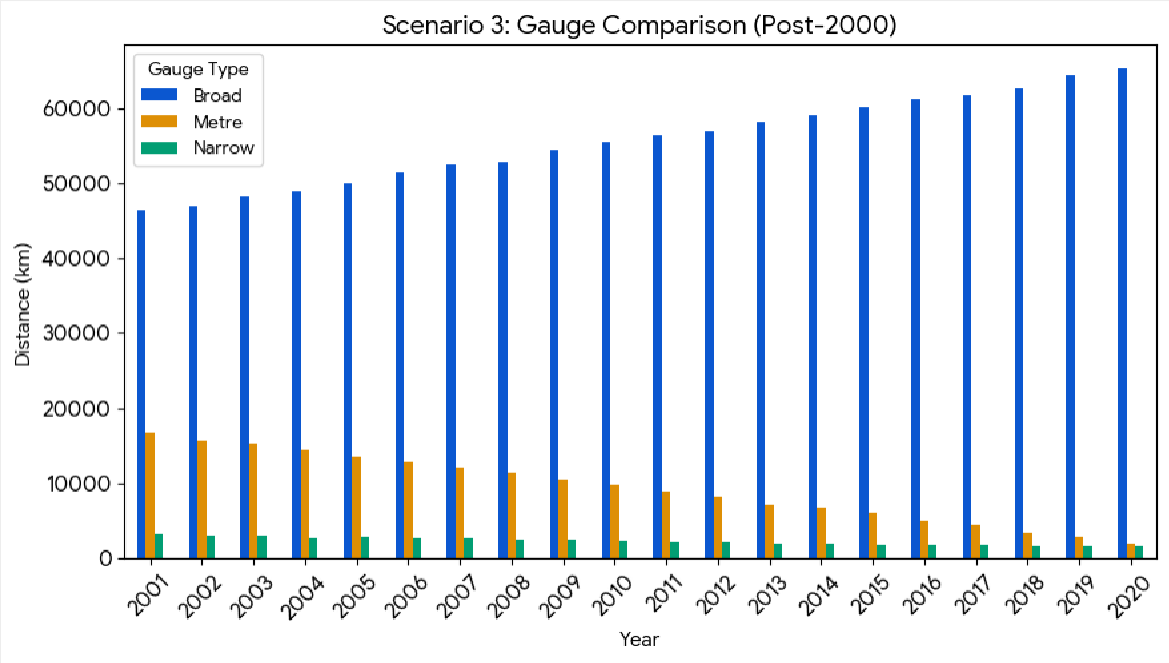
Scenario 2: Simple Visualization

Total railway track growth is visualized below. **Trend:** The trend is generally increasing as seen in the visualization.



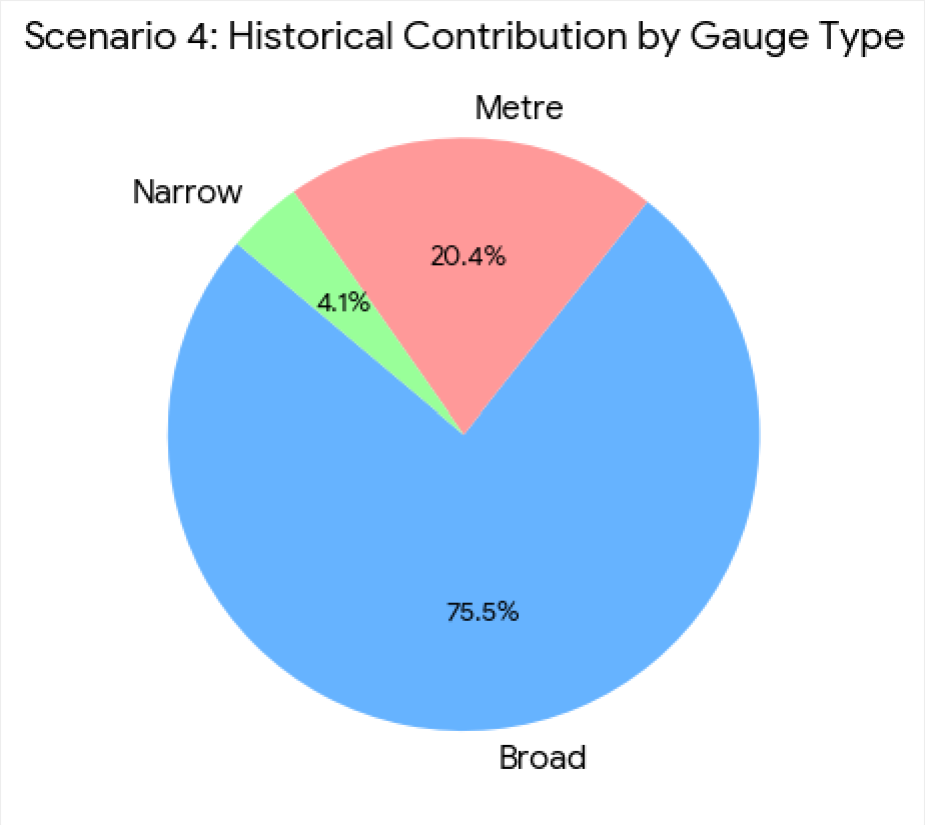
Scenario 3: Filtering + Bar Chart

Analyzing modern expansion (post-2000). **Dominant Gauge:** Broad Gauge has clearly dominated infrastructure expansion in recent years.



Scenario 4: Feature Engineering + Pie Chart

Historical contribution of each gauge type based on total sum of track length across all years.



Broad Gauge contributes the largest share ($> 70\%$) of the total track distance recorded.

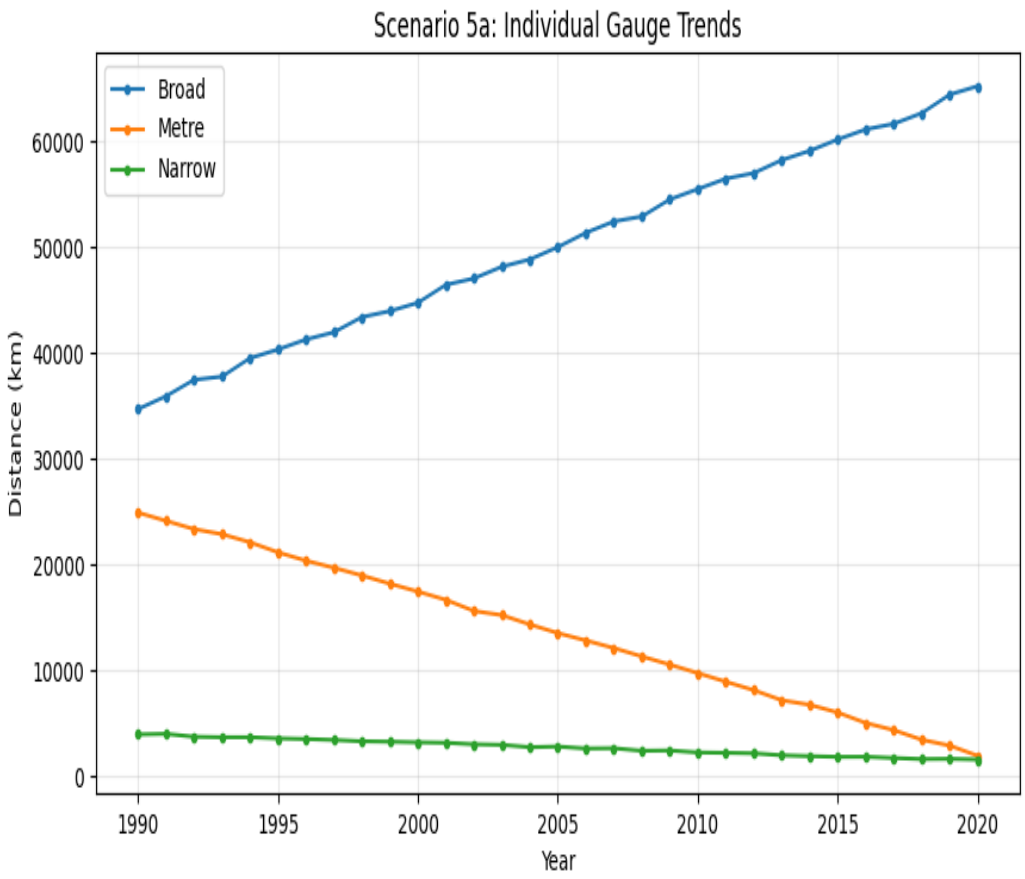
Scenario 5: Advanced Analysis + Multiple Graphs

- Percentage composition and yearly growth calculations using NumPy.

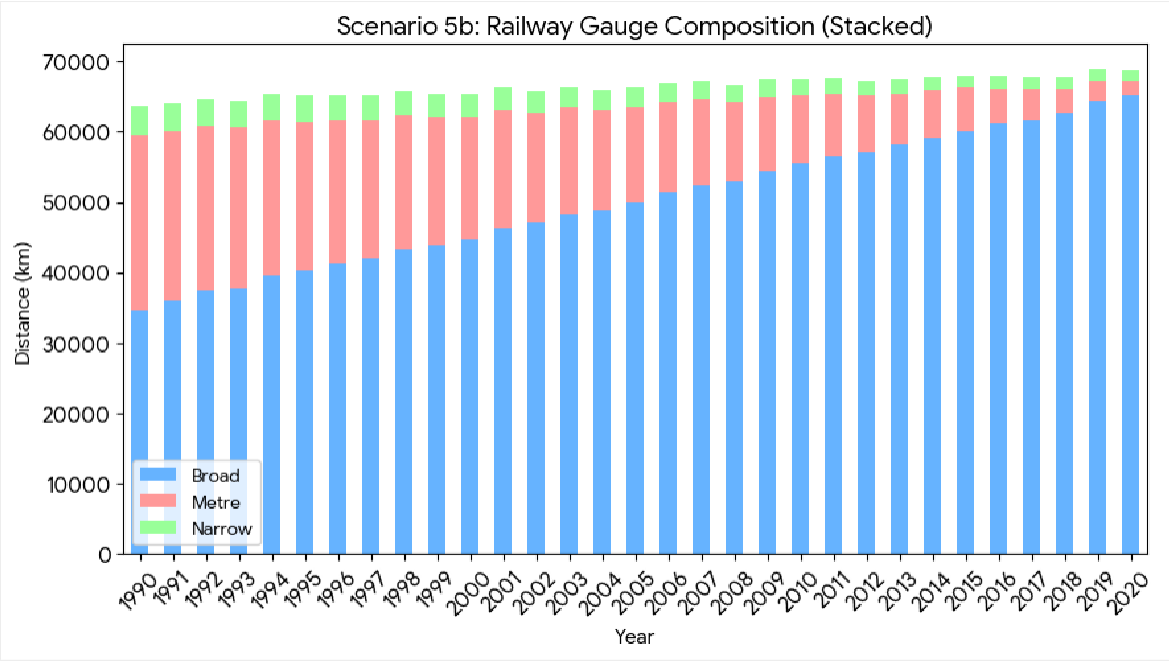
Highlights:

- **Year with highest growth:** 2019
- **Decline in Metre Gauge:** Yes
- **Decline in Narrow Gauge:** Yes

Individual Gauge Trends



Railway Gauge Composition (Stacked)



Final Conclusion:

"Is the railway system shifting towards a single dominant gauge?"

Yes. The visualizations clearly show that while the total network grows, Broad Gauge is expanding rapidly while Metre and Narrow Gauges are being phased out, confirming a move toward a single dominant gauge (Standardization).